

Question Paper

Exam Date & Time: 24-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

FIRST SEMESTER M.TECH. DEGREE MAKEUP EXAMINATION - DECEMBER 2025

Data Structures and Algorithms [CSS_5114]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

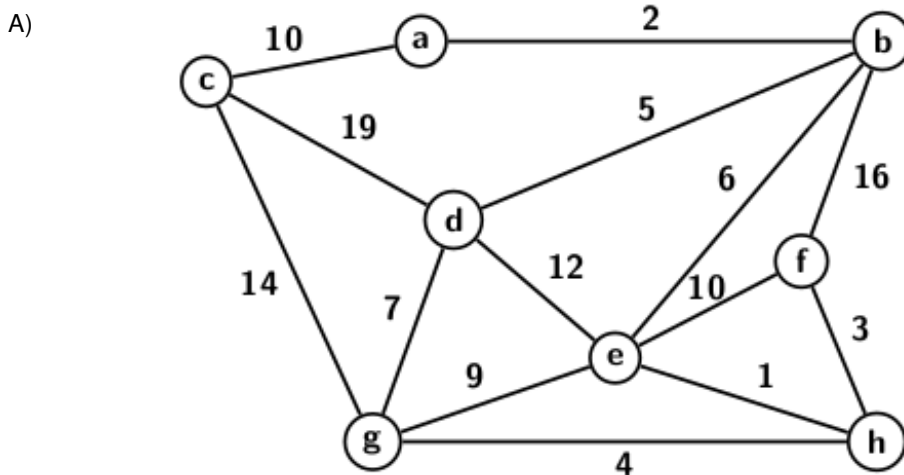
- 1) Consider the following sequence of insertions into an AVL tree: (5)
80, 60, 90, 50, 100, 40, 120, 30, 70, 110
- A) Draw the AVL tree after each insertion and show the required rotations.
Explain the balance factor after each insertion and rotation.
Course Outcome: "CO4" / Cognitive Level: "Application"
- B) Implement a **queue using two stacks**. Perform the following operations: (3)
 - **Enqueue** the values: **5, 10, 15, 20, 25**.
 - **Dequeue** twice and then **Enqueue: 30, 35, 40**.**Dequeue** two more values and show the final state of the queue.
Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Define an array of **5 student structures** where each structure contains **name** (string), **roll number** (integer), and **marks** (integer array of size 3). (2)
Write a program to calculate the average marks of all students using pointers.
Course Outcome: "CO3" / Cognitive Level: "Application"
- 2) Convert the following infix expression to **postfix**: (5)
(A + B) * (C - D) / (E + F) - (G + H)
- A)
 - Show all the steps involved in the conversion using a stack.
Explain how to handle the operators and parentheses during the conversion process.
Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Write an algorithm to check if the following expression has balanced parentheses: (3)
{[(a + b) * (c / d)] - (e + f)}
Show the step-by-step execution of the algorithm with an explanation of how the stack is used to match the parentheses.
Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Define a structure for a **book** containing **title**, **author**, and **publisher**. (2)
Inside the book structure, define another structure for **publisher details** containing **name** and **address**.

- 3) Sort the following list using **Merge Sort**:
88, 12, 25, 34, 55, 15, 62, 90, 53, 73 (5)
- A) Show the recursive splitting and merging process. Write a function to perform the merge operation.

- B) Implement **Binary Search** for element **42** in the sorted array:
arr[10] = {12, 14, 25, 32, 36, 42, 55, 66, 72, 88} (3)
- Develop the same with iterative method

- C) Given the graph with edges:
A → B = 3, B → C = 5, C → D = 2, D → A = 4 (2)
- Compute the **total weight of the edges** and check if the graph is **connected**.

- 4) Given the following graph with **nodes** and **weighted edges**: (5)



Find the **Minimum Spanning Tree (MST)** of the given graph using **Kruskal's Algorithm**.

- Show the steps involved in constructing the MST, including the edges chosen and their weights.

Explain how the algorithm works step-by-step.

- B) Create a **circular linked list** and perform the following operations: (3)
- Insert nodes with values: 10, 20, 30, 40, 50.**

- Insert a node at the beginning, at the end, and at position 3.
- Display the final list after the operations.

Develop the code for the same.

- C) **Analyze the time complexity of the following code snippet using Big O notation** (2)

```

for(i = 0; i < n; i++)
    for(j = 0; j < n; j++)
        for(k = j; k < n; k++)
            printf("%d", i + j + k);

```

Course Outcome: "CO3" / Cognitive Level: "Application"

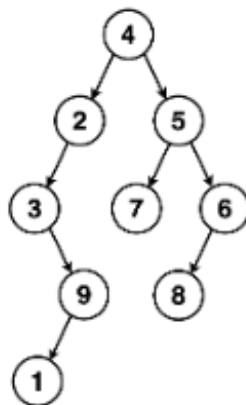
- 5) Given the infix expression: $A + (B * C) - (D/E)$, (5)
 construct the corresponding expression tree.
- A) Explain each step in the construction of the tree. Construct
 In-order, pre-order and post-order traversals on the same.
 Replace $A=9$, $B=3$, $C=2$, $D=4$, $E=1$. Compute the equivalent
 value of expression using post-order traversal.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Sort the following array using the **Bubble Sort** algorithm: (3)
10, 7, 8, 9, 1, 5, 3, 4, 6
 Show the array after each pass of the algorithm.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Given the following binary tree: (2)



Convert this tree into a Binary Search Tree (BST) and
 explain the conversion process.

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM PG-2025-
 2026

CSS_5114:Data Structures and Algorithms

#	Course Outcome	Description	Marks
1	1	CO 1	11
2	2	CO 2	11
3	3	CO 3	9
4	4	CO 4	12
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	15	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)